

GUILLERMO FERRIOL RAVELO

Full-Stack Software Engineer — Web & Mobile

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SUMMARY

Software engineer with 3+ years shipping production mobile and web products across mobility, IoT access control, insurtech, retail, and B2B supply. Specialized in cross-platform mobile (Flutter/Dart) and modern web (TypeScript, React, Next.js, NestJS, Node.js), with hands-on work in real-time geolocation and fleet tracking, RFID/IoT integration, microfrontend architecture, and machine-learning integration. I own features end to end: system design, scalable back ends (Go, FastAPI, GraphQL), and clean, maintainable UIs. I also work with AI coding agents (Claude Code, Codex, GitHub Copilot, OpenCode), applying spec-driven development, harness engineering, and loop engineering to ship faster with higher quality.

EXPERIENCE

Pyxel Solutions — Mobile / Full-Stack Developer

2025 – Present

La Habana, Cuba

- Develop cross-platform mobile applications in Flutter, owning features end to end from UI to back-end integration, shipping 4+ production apps spanning real-time mobility, on-demand services, POS, and ERP.
- Build real-time geolocation, live-maps, and driver/rider matching features and integrate GraphQL/REST back ends and third-party map services, delivering low-latency tracking and updates, by designing efficient data flows and state management on mobile.

Countigo — Frontend Developer

2024 – 2025

La Habana, Cuba

- Architected a microfrontend platform (Module Federation) for conditional-TV products, enabling 3+ independently deployable modules and cutting release-coordination time by ~25%, by splitting a monolith into modular React/Next.js apps.
- Shipped multi-tenant management apps for conditional-TV stores and channels, serving 5+ entities/channels, by building reusable, modular UI with React/Next.js, Ant Design, Zustand, and Redux.

Bufetes Colectivos (ONBC) — Full-Stack Developer

2023 – 2024

La Habana, Cuba

- Delivered a full-stack legal case & event management system (React/Next.js front end, Go/Fiber + PostgreSQL back end) that centralized firm operations, reducing manual processing time by ~30%, by automating core administrative workflows.
- Integrated a machine-learning model into the firm's workflow, automating a key classification task over thousands of records, by building the data pipeline and model-serving layer.

PROJECTS

Mobility & Real-Time

- FAIND** — Ride-hailing platform (Uber-style): real-time driver/rider matching, live geolocation, and multi-operator fleet control. *Flutter, GraphQL, real-time maps.*
- Trackmont** — Satellite GPS fleet & asset tracking: live fleet map, instant alerts, and data-driven telematics, all from mobile. *Flutter, Node.js, maps, real-time location streaming.*

IoT & Access Control

- SEGUIPASS** — Smart access control for gated communities: RFID-TAG vehicle authorization, QR visitor passes, real-time entry/exit notifications, access history, and misuse reporting. *Flutter + RFID/IoT integration.*

Insurtech & On-Demand

- DelToro Insurance** — Bilingual (EN/ES) insurance platform: instant free quotes (auto, health, home, life), agent connection, and lead capture. *Flutter, internationalization.*
- Charge2Go** — On-demand EV-charging app: request a mobile charger to your location. *Flutter, geolocation.*

Business Systems

- ArgoPos** — Point-of-sale (POS) app: sales, checkout, and inventory. *Flutter.*
- MHHome** — All-in-one home-improvement ERP app covering interior and exterior projects, from protection to design. *Flutter.*

- **Decorect (web)** — Premium tile-design supplier: online catalog, live inventory, and online ordering. *React / Next.js, NestJS.*
- **Ardent (web)** — B2B distribution platform for hotels & restaurants: design, store, and deliver branded essentials. *React / Next.js, NestJS.*

Broadcast / Conditional TV

- **Suite TV (web)** — Management system for a conditional-TV store and its products, built on a microfrontend architecture. *React / Next.js, Module Federation.*
- **Count TV (web)** — Multi-tenant management system for conditional-TV channels; multiple entities manage and customize internal channels. *React / Next.js.*

More projects (ML, systems, data) on GitHub — github.com/guilleferrioldev

SKILLS

Languages	TypeScript, Python, Go, Dart, Bash
Mobile	Flutter, real-time geolocation & maps, RFID/IoT integration
Frontend	React, Next.js, Module Federation (microfrontends), Redux, Zustand, Tailwind CSS, Ant Design, Three.js, GSAP
Backend	Node.js, NestJS, Go (Fiber, gRPC, NATS), FastAPI, REST & GraphQL APIs
Data / ML	Scikit-learn, Pandas, NumPy, Matplotlib, Plotly, Streamlit
Databases	PostgreSQL, SQLite, Cassandra, ScyllaDB
Infra / Tools	Docker, GitHub Actions (CI/CD), Vercel, Kafka, RabbitMQ, Git, Postman, Linux (Arch)
Testing	Automated testing (unit & integration)
Engineering	System Design, Software Architecture, Real-Time Systems, Design Patterns, SOLID/KISS/YAGNI/GRASP, Data Structures & Algorithms, OOP / Functional Programming

AI-ASSISTED DEVELOPMENT

Hands-on orchestrating AI coding agents in production workflows — not just prompting, but engineering the systems around them.

Tools: Claude Code, Codex, GitHub Copilot, OpenCode, Ollama (local LLMs)

- **Spec-Driven Development** — treat a structured spec as the single source of truth and drive the agent to generate, test, and validate against it; focus on intent and acceptance criteria over manual coding.
- **Harness Engineering** — design the environment, constraints, and feedback loops around an agent (Agent = Model + Harness) so it ships reliable code at scale with less review toil.
- **Loop Engineering** — design autonomous act → observe → decide → repeat loops (ReAct pattern) with clear goals and stopping conditions so agents iterate and self-correct toward a target.

EDUCATION

University of Havana (MATCOM) <i>Data Science</i>	2022 – 2024
Tech. University of Havana (CUJAE) <i>Architecture</i>	2019 – 2022

LANGUAGES

Spanish (native)
English (professional working proficiency)